

SWAMI SANT DASS PUBLIC SCHOOL

Product Inventory Management

Group 8 (D)

SUBMITTED BY

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SUBMITTED TO

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PGT Computer Science

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1. PROJECT OVERVIEW & SPECIFICATIONS

1.1 Objective

The objective of this project is to develop a console based inventory management application in Python using dictionaries, loops, conditional statements, and lists. The system maintains product records indexed by unique Product Codes and provides structured operations for data insertion, display, search, modification, and query filtering.

1.2 Scope and Methodology

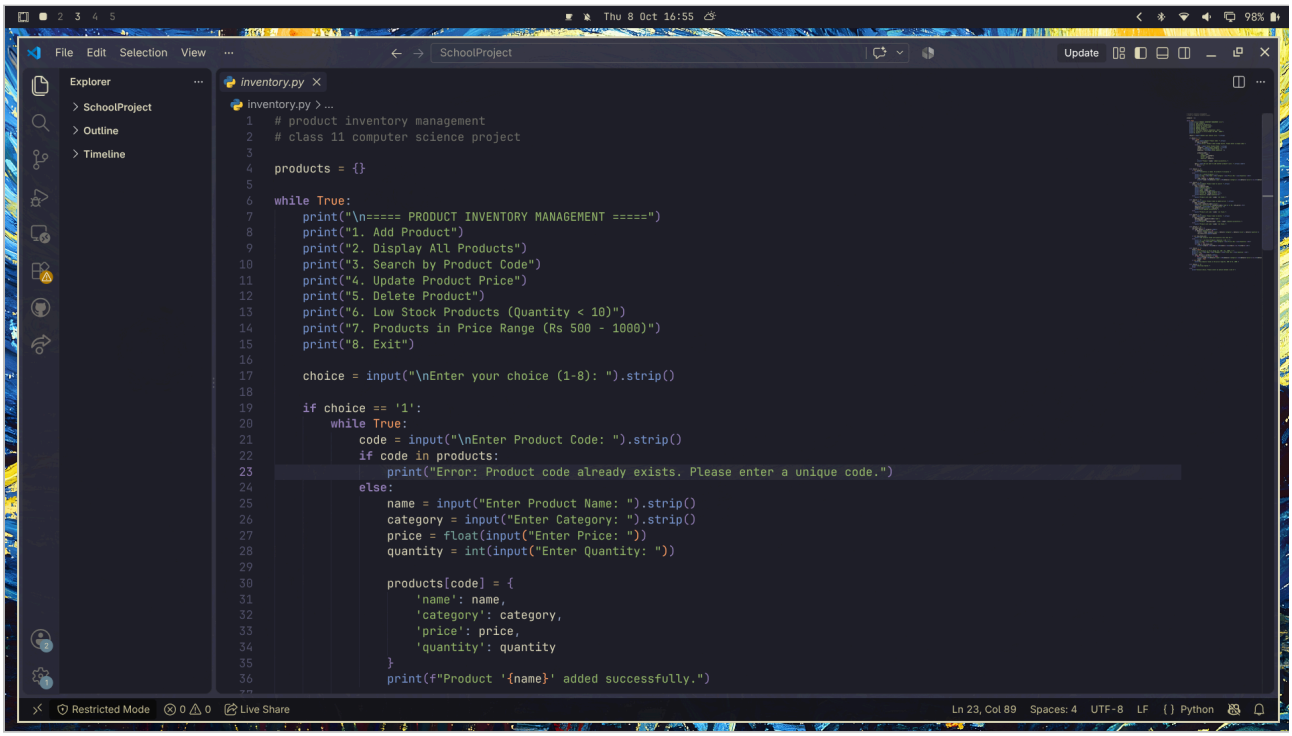
The project is developed using pure procedural logic in accordance with syllabus guidelines. In-memory data management is handled through a nested dictionary representation where each record stores the product name, category, unit price, and current quantity.

1.3 Functional Requirements

1. **Add Products:** Accept product details with continuous entry until terminated by the user, enforcing unique product codes.
2. **Display Products:** Tabular display of the complete product inventory with all attributes.
3. **Search by Product Code:** Search and display individual records, reporting appropriate messages if not found.
4. **Update Price:** Locate an existing product code and modify only its unit price.
5. **Delete Product:** Validate existence and remove the corresponding record from the dictionary.
6. **Low Stock List:** Extract records with quantity under 10 into a separate list structure and present the results.
7. **Products in Price Range:** Filter and display products priced between Rs. 500 and Rs. 1000 inclusive.

2. PYTHON SOURCE CODE (PART 1 OF 2)

Lines 1 - 36: Initialization, Main Loop, and Product Insertion



```python
# product inventory management
# class 11 computer science project

products = {}

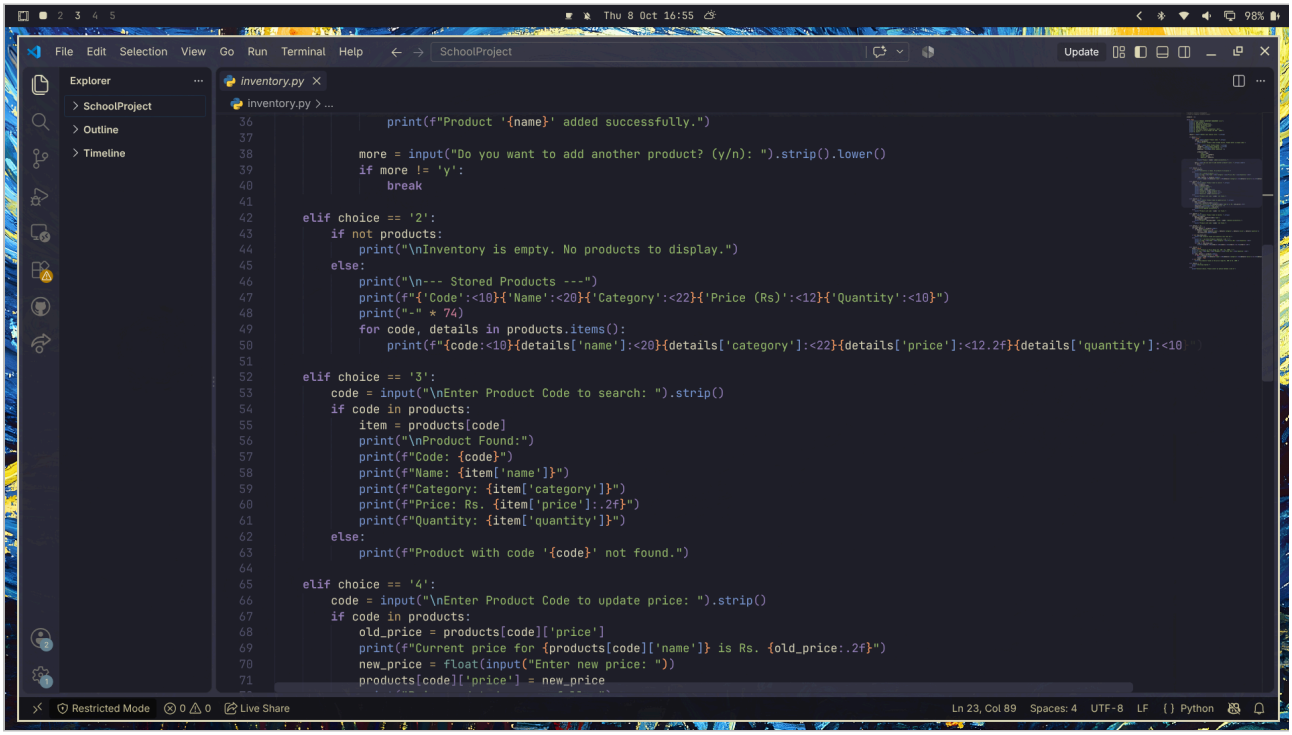
while True:
    print("\n===== PRODUCT INVENTORY MANAGEMENT =====")
    print("1. Add Product")
    print("2. Display All Products")
    print("3. Search by Product Code")
    print("4. Update Product Price")
    print("5. Delete Product")
    print("6. Low Stock Products (Quantity < 10)")
    print("7. Products in Price Range (Rs 500 - 1000)")
    print("8. Exit")

    choice = input("\nEnter your choice (1-8): ").strip()

    if choice == '1':
        while True:
            code = input("\nEnter Product Code: ").strip()
            if code in products:
                print("Error: Product code already exists. Please enter a unique code.")
            else:
                name = input("Enter Product Name: ").strip()
                category = input("Enter Category: ").strip()
                price = float(input("Enter Price: "))
                quantity = int(input("Enter Quantity: "))

                products[code] = {
                    'name': name,
                    'category': category,
                    'price': price,
                    'quantity': quantity
                }
                print(f"Product '{name}' added successfully.")
```

Lines 36 - 71: Display Catalog, Code Search, and Price Update



```python
                print(f"Product '{name}' added successfully.")

            more = input("Do you want to add another product? (y/n): ").strip().lower()
            if more != 'y':
                break

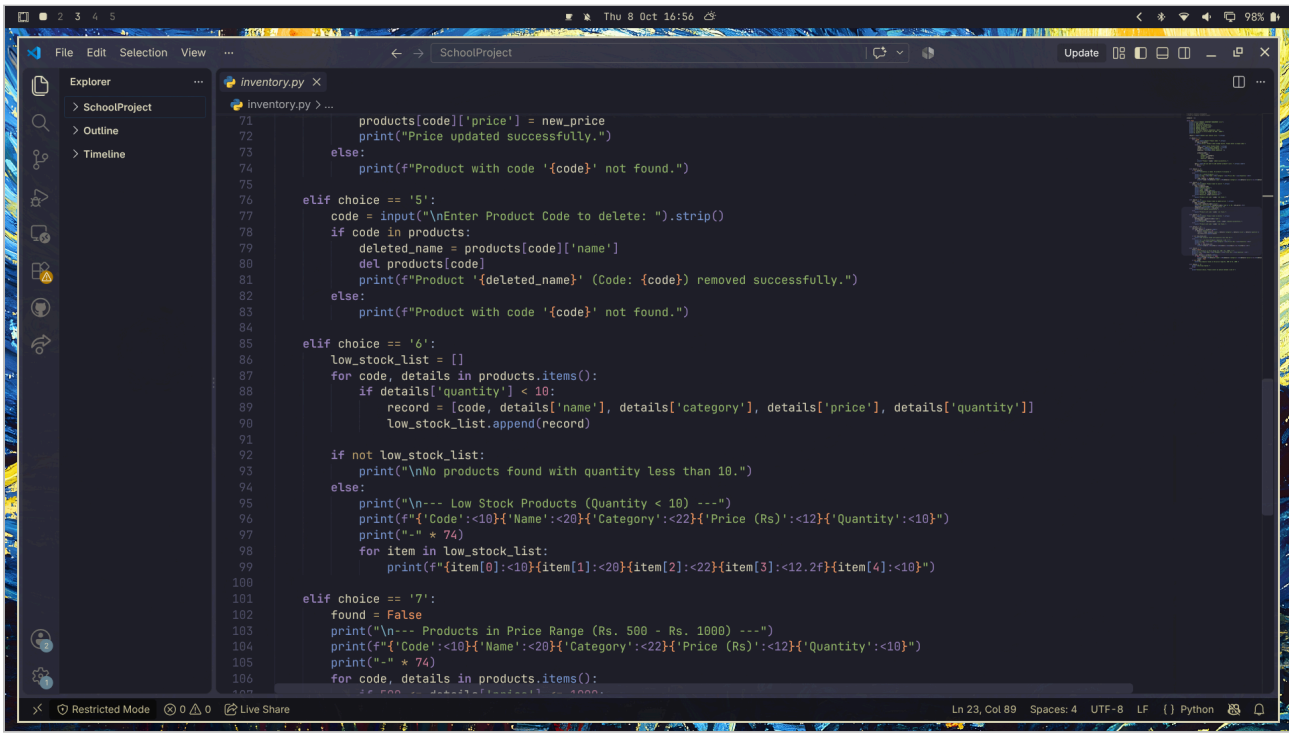
    elif choice == '2':
        if not products:
            print("\nInventory is empty. No products to display.")
        else:
            print("\n--- Stored Products ---")
            print(f"{'Code':<10}{'Name':<20}{'Category':<22}{'Price (Rs)':<12}{'Quantity':<10}")
            print("-" * 74)
            for code, details in products.items():
                print(f"{code:<10}{details['name']:<20}{details['category']:<22}{details['price']:<12.2f}{details['quantity']:<10}")

    elif choice == '3':
        code = input("\nEnter Product Code to search: ").strip()
        if code in products:
            item = products[code]
            print("\nProduct Found:")
            print(f"Code: {code}")
            print(f"Name: {item['name']}")
            print(f"Category: {item['category']}")
            print(f"Price: Rs. {item['price']:.2f}")
            print(f"Quantity: {item['quantity']}")
        else:
            print(f"Product with code '{code}' not found.")

    elif choice == '4':
        code = input("\nEnter Product Code to update price: ").strip()
        if code in products:
            old_price = products[code]['price']
            print(f"Current price for {products[code]['name']} is Rs. {old_price:.2f}")
            new_price = float(input("Enter new price: "))
            products[code]['price'] = new_price
```

2. PYTHON SOURCE CODE (PART 2 OF 2)

Lines 71 - 107: Product Deletion and Low Stock Extraction



```
            products[code]['price'] = new_price
            print("Price updated successfully.")
        else:
            print(f"Product with code '{code}' not found.")

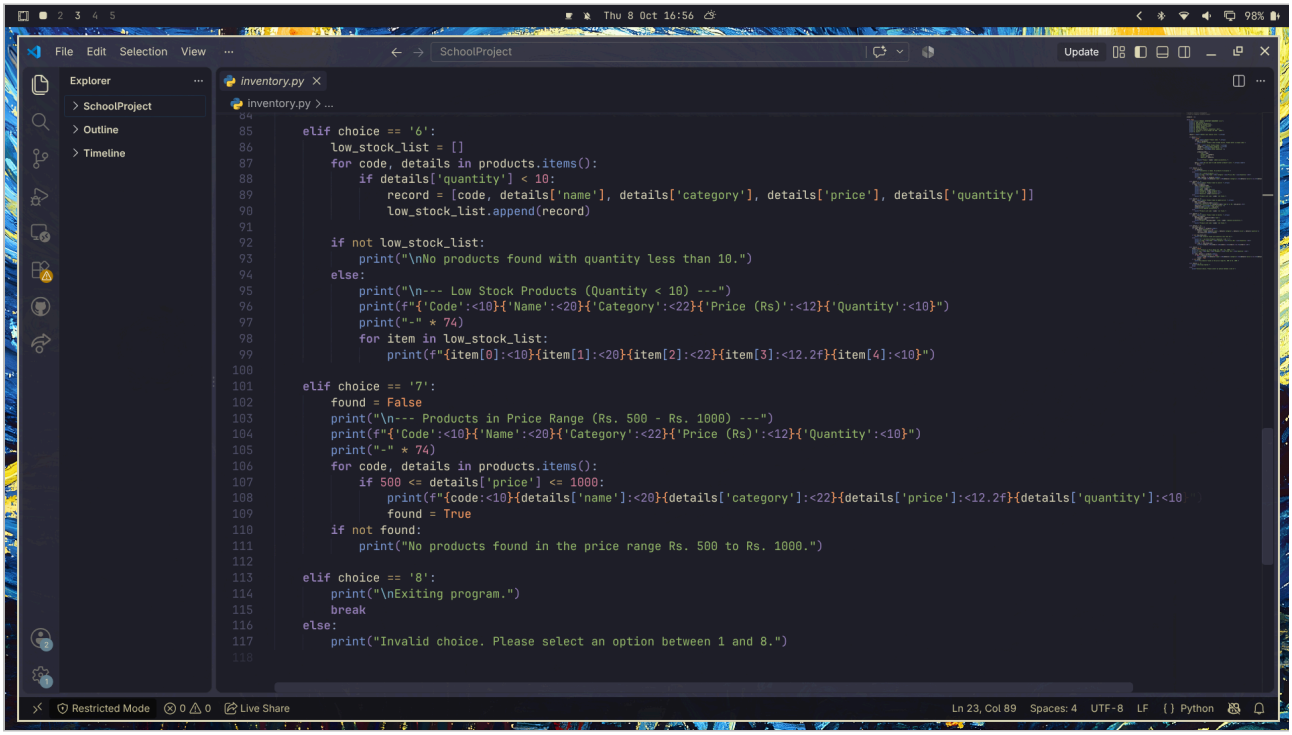
    elif choice == '5':
        code = input("\nEnter Product Code to delete: ").strip()
        if code in products:
            deleted_name = products[code]['name']
            del products[code]
            print(f"Product '{deleted_name}' (Code: {code}) removed successfully.")
        else:
            print(f"Product with code '{code}' not found.")

    elif choice == '6':
        low_stock_list = []
        for code, details in products.items():
            if details['quantity'] < 10:
                record = [code, details['name'], details['category'], details['price'], details['quantity']]
                low_stock_list.append(record)

        if not low_stock_list:
            print("\nNo products found with quantity less than 10.")
        else:
            print("\n--- Low Stock Products (Quantity < 10) ---")
            print(f"{'Code':<10}{'Name':<20}{'Category':<22}{'Price (Rs)':<12}{'Quantity':<10}")
            print("-" * 74)
            for item in low_stock_list:
                print(f"{item[0]:<10}{item[1]:<20}{item[2]:<22}{item[3]:<12.2f}{item[4]:<10}")

    elif choice == '7':
        found = False
        print("\n--- Products in Price Range (Rs. 500 - Rs. 1000) ---")
        print(f"{'Code':<10}{'Name':<20}{'Category':<22}{'Price (Rs)':<12}{'Quantity':<10}")
        print("-" * 74)
        for code, details in products.items():
```

Lines 84 - 118: Price Range Filter and Exit Routine



```
    elif choice == '6':
        low_stock_list = []
        for code, details in products.items():
            if details['quantity'] < 10:
                record = [code, details['name'], details['category'], details['price'], details['quantity']]
                low_stock_list.append(record)

        if not low_stock_list:
            print("\nNo products found with quantity less than 10.")
        else:
            print("\n--- Low Stock Products (Quantity < 10) ---")
            print(f"{'Code':<10}{'Name':<20}{'Category':<22}{'Price (Rs)':<12}{'Quantity':<10}")
            print("-" * 74)
            for item in low_stock_list:
                print(f"{item[0]:<10}{item[1]:<20}{item[2]:<22}{item[3]:<12.2f}{item[4]:<10}")

    elif choice == '7':
        found = False
        print("\n--- Products in Price Range (Rs. 500 - Rs. 1000) ---")
        print(f"{'Code':<10}{'Name':<20}{'Category':<22}{'Price (Rs)':<12}{'Quantity':<10}")
        print("-" * 74)
        for code, details in products.items():
            if 500 <= details['price'] <= 1000:
                print(f"{code:<10}{details['name']:<20}{details['category']:<22}{details['price']:<12.2f}{details['quantity']:<10
                found = True
        if not found:
            print("No products found in the price range Rs. 500 to Rs. 1000.")

    elif choice == '8':
        print("\nExiting program.")
        break
    else:
        print("Invalid choice. Please select an option between 1 and 8.")
```

3. TEST RUN & OUTPUT EVIDENCE

Terminal execution transcript demonstrating all 7 operations and validation rules:

```
===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 1

Enter Product Code: 101
Enter Product Name: Keyboard
Enter Category: Computer Accessory
Enter Price: 750
Enter Quantity: 15
Product 'Keyboard' added successfully.
Do you want to add another product? (y/n): y

Enter Product Code: 102
Enter Product Name: Mouse
Enter Category: Computer Accessory
Enter Price: 550
Enter Quantity: 8
Product 'Mouse' added successfully.
Do you want to add another product? (y/n): y

Enter Product Code: 103
Enter Product Name: USB Cable
Enter Category: Electronics
Enter Price: 250
Enter Quantity: 20
Product 'USB Cable' added successfully.
Do you want to add another product? (y/n): y

Enter Product Code: 104
Enter Product Name: Webcam
Enter Category: Computer Accessory
Enter Price: 1200
Enter Quantity: 5
Product 'Webcam' added successfully.
Do you want to add another product? (y/n): y

Enter Product Code: 105
Enter Product Name: Laptop Stand
Enter Category: Office
Enter Price: 850
Enter Quantity: 12
Product 'Laptop Stand' added successfully.
Do you want to add another product? (y/n): y

Enter Product Code: 101
Error: Product code already exists. Please enter a unique code.
Do you want to add another product? (y/n): n

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 2

--- Stored Products ---
Code      Name               Category             Price (Rs)  Quantity
-------------------------------------------------------------------------
101       Keyboard           Computer Accessory    750.00      15
102       Mouse              Computer Accessory    550.00      8
```

```
103       USB Cable           Electronics           250.00      20
104       Webcam              Computer Accessory    1200.00     5
105       Laptop Stand        Office                850.00      12

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 3

Enter Product Code to search: 102

Product Found:
Code: 102
Name: Mouse
Category: Computer Accessory
Price: Rs. 550.00
Quantity: 8

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 3

Enter Product Code to search: 999
Product with code '999' not found.

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 4

Enter Product Code to update price: 102
Current price for Mouse is Rs. 550.00
Enter new price: 580
Price updated successfully.

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 5

Enter Product Code to delete: 103
Product 'USB Cable' (Code: 103) removed successfully.

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
```

```
8. Exit

Enter your choice (1-8): 6

--- Low Stock Products (Quantity < 10) ---
Code      Name                Category              Price (Rs)  Quantity
--------------------------------------------------------------------------
102       Mouse               Computer Accessory    580.00      8
104       Webcam              Computer Accessory    1200.00     5

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 7

--- Products in Price Range (Rs. 500 - Rs. 1000) ---
Code      Name                Category              Price (Rs)  Quantity
--------------------------------------------------------------------------
101       Keyboard            Computer Accessory    750.00      15
102       Mouse               Computer Accessory    580.00      8
105       Laptop Stand        Office                850.00      12

===== PRODUCT INVENTORY MANAGEMENT =====
1. Add Product
2. Display All Products
3. Search by Product Code
4. Update Product Price
5. Delete Product
6. Low Stock Products (Quantity < 10)
7. Products in Price Range (Rs 500 - 1000)
8. Exit

Enter your choice (1-8): 8

Exiting program.
```